Weighted L^1 interchange for the log moments

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Abstract

The paper's fluctuation coefficients are series indexed by tree types, integrated against the Gaussian log weight $w(s) = s^{\alpha-1}(\log s)^{\ell} e^{-\beta s^2}$. This unit exposes the interchange API for such series: if the weighted L^1 norms $\int |w f_t|$ are summable then $\mathcal{M}(\sum_t f_t) = \sum_t \mathcal{M}(f_t)$ with absolute convergence (`hasSum_logMoment`); a normal-convergence hypothesis $\sum_t |f_t(s)| \leq G(s)$ with $|w|G$ integrable supplies both hypotheses through a Tonelli argument in `ENNReal` (`weighted_summable_of_dominated`); and the envelope $C(1+s)^D e^{\beta s a'}$ is integrable against $|w|$ for $\alpha > 0$ (`hasSum_logMoment_of_envelope`). This is Astra #4's rank-3 unit and the tool for the first genuinely infinite coefficient identity (the axis series for the $d = 2$ constant coefficient B). Zero `sorry`/`axiom`.

1 The things formalised

```
noncomputable def momentWeight (   :  ) (  :  ) (s :  ) :   :=
  s ^ (  - 1) * Real.log s ^   * Real.exp (-  * s ^ 2)
theorem logMoment_eq_integral_weight (   :  ) (  :  ) (c :   →  ) :
    logMoment      c =   s in Ioi (0 :  ), momentWeight      s * c s
theorem hasSum_logMoment {  : Type*} [Countable  ] (   :  ) (  :  ) (f :   →   →  )
    (hint :   t, IntegrableOn (fun s => momentWeight      s * f t s) (Ioi 0))
    (hsum : Summable fun t =>   s in Ioi (0 :  ), |momentWeight      s * f t s|) :
    HasSum (fun t => logMoment      (f t)) (logMoment      (fun s =>  ' t, f t s))
theorem logMoment_tsum ... :
    logMoment      (fun s =>  ' t, f t s) =  ' t, logMoment      (f t)
theorem weighted_summable_of_dominated {  : Type*} [Countable  ] (w :   →  )
    (hw : Measurable w) (f :   →   →  ) (hf :   t, Measurable (f t)) (G :   →  )
    (hsumm :   s   Ioi (0 :  ), Summable fun t => |f t s|)
    (hdom :   s   Ioi (0 :  ),  ' t, |f t s|   G s)
    (hG : IntegrableOn (fun s => |w s| * G s) (Ioi 0)) :
    (  t, IntegrableOn (fun s => w s * f t s) (Ioi 0))
      Summable fun t =>   s in Ioi (0 :  ), |w s * f t s|
theorem hasSum_logMoment_of_dominated ...
theorem momentWeight_envelope_integrableOn (  a'   C :  ) (h  : 0 <  ) (h  : 0 <  )
    (hC : 0   C) (  D :  ) :
    IntegrableOn
      (fun s => |momentWeight      s| * (C * (1 + s) ^ D * Real.exp (  * s * a'))) (Ioi 0)
theorem hasSum_logMoment_of_envelope {  : Type*} [Countable  ] (  a'   C :  ) (h  : 0 <  )
    (h  : 0 <  ) (hC : 0   C) (  D :  ) (f :   →   →  ) (hf :   t, Measurable (f t))
    (hsumm :   s   Ioi (0 :  ), Summable fun t => |f t s|)
    (hdom :   s   Ioi (0 :  ),  ' t, |f t s|   C * (1 + s) ^ D * Real.exp (  * s * a')) :
    HasSum (fun t => logMoment      (f t)) (logMoment      (fun s =>  ' t, f t s))
```

2 Mathematics

The core is Mathlib's `hasSum_integral_of_summable_integral_norm`: for integrable F_t with $\sum_t \|F_t\|_{L^1} < \infty$, $\sum_t \int F_t = \int \sum_t F_t$ as a `HasSum`. With $F_t = wf_t$ the left side is $\sum_t \mathcal{M}(f_t)$ and the integrand on the right is $w \sum_t f_t$ by `tsum_mul_left` (unconditional in a division ring).

The domination step converts $\sum_t |f_t| \leq G$ into the two hypotheses by working in `ENNReal`: $\sum_t \int^- \|wf_t\|_e = \int^- \sum_t \|wf_t\|_e$ (`lintegral_tsum`, no summability needed) and pointwise $\sum_t \|wf_t\|_e = \mathrm{ofReal}|w| \cdot \mathrm{ofReal} \sum_t |f_t| \leq \| \, |w|G\|_e$ (`ENNReal.ofReal_tsum_of_nonneg` uses the pointwise summability). Finiteness of $\int^- \| |w|G\|_e$ is the `HasFiniteIntegral` half of the integrability of $|w|G$. Then each $\int^- \|wf_t\|_e$ is finite (integrability) and $\int |wf_t| = (\int^- \|wf_t\|_e)_{\mathrm{toReal}}$ is summable by `ENNReal.summable_toReal`.

For the envelope, $|w(s)| \, C(1+s)^D e^{\beta s a'} \leq s^{\alpha-1}(1+|\log s|)^\ell e^{-\beta s^2} \cdot C(1+s)^D e^{\beta s a'}$, integrable by unit 96's `moment_integrableOn_of_envelope` for $\alpha > 0$.

3 Paper-facing meaning

This is the interchange lemma Astra #3(c) asked for, in the form recommended in Astra #4(c): an arbitrary measurable weight first, the Gaussian log weight as a specialisation, with the normal-convergence corollary. Its first application will be the absolutely convergent series for the axis integral $I_\Psi(s) = \sum_{i \geq 1} \Psi_i(s) b^i / i$ and hence for the $d = 2$ constant coefficient B when Φ has convergent Taylor series on a strictly larger disc. No convergence of the inverse-N expansion is implied.

4 Lean notes

- `hasSum_integral_of_summable_integral_norm` with explicit `(  := volume.restrict (Ioi 0))` left `NormedSpace   ?E` stuck (the `0` in `Ioi 0` has no type there). Give the `have` an explicit `HasSum` type and let unification fill everything.

- `Measurable.abs` does not exist; use `continuous_abs.measurable.comp`.

- `Real.enorm_eq_ofReal_abs`, `ENNReal.ofReal_mul`, `ENNReal.tsum_mul_left`, `ENNReal.ofReal_tsum_of_nonneg` chain the pointwise `ENNReal` identity; `gcongr` handles the final `ofReal` monotonicity.

- `integral_norm_eq_lintegral_enorm` converts the real L^1 norm to the `toReal` of the lower integral, matching `ENNReal.summable_toReal`.

- An envelope coefficient must be assumed nonnegative (`hC : 0   C`) for `positivity`; a bound of the form $|c| \leq A(\dots)$ with $A < 0$ is vacuous anyway.